WAB

Provadis School of International Management and Technology

Examining the Feasibility of Machine Learning-Based LogP Prediction on an ESP32 Microcontroller

A Proof-of-Concept Implementation and Performance Analysis

Rubin Chempananickal James

rubin.chempananickal-james@stud-provadis-hochschule.de

Matriculation Number: D876

Department: Information Technology

Module: Embedded Systems and Software

Reviewer: Prof. Dr. Eric Hutter

March 29, 2026

Abstract

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AI Declaration

The usage of AI tools within this project is documented here. I solemnly declare that I have documented all interactions with AI tools, including the prompts used and the outputs received.

System	Prompt	Usage
GitHub Copilot 1	Asked to delete the existing project and provide a simple LaTeX template with a shared preamble, a Chapters subdirectory, TOC, glossary, abbreviations, and Roman numerals for non-content pages.	Template structure and LaTeX setup provided
GitHub Copilot 2	Requested additional title page lines (WAB header, reviewer, module) for the main document.	Title page metadata and layout updated
GitHub Copilot 3	Requested uppercase Roman numerals for front matter and lowercase Roman numerals for back matter page numbering.	Page numbering adjusted in main and exposee
GitHub Copilot 4	Requested exposee title page to include shared info fields (WAB header, department, module, reviewer).	Exposee title page updated to include shared metadata
GitHub Copilot 5	Requested adding the provadis-hochschule.pdf logo to the top right corner of both main and exposee title pages.	Logo added to top right of both title pages
GitHub Copilot 6	Reported that Glossary and Abbreviations sections were missing from compiled output.	Added example `\newacronym` entries to abbreviations.tex so the Abbreviations section displays properly
GitHub Copilot 7	Reported that glossary section still not appearing in final PDF.	Updated settings.tex to include `automake` option in glossaries package to enable automatic glossary generation
GitHub Copilot 8	Asked for boilerplate to train a TFLite model for ESP32 to estimate logP from SMILES.	Provided a Python training script (RDKit + Keras) with TFLite export

System	Prompt	Usage
GitHub Copilot 9	Asked how to run inference on an ESP32 with the trained model.	Provided a high-level TFLite Micro deployment workflow for ESP32
GitHub Copilot 10	Asked for a beginner-friendly README in Code/ with step-by-step setup and usage.	Added a novice-oriented Code/README.md with installation, training, and ESP32 notes
GitHub Copilot 11	Asked for a TFLite Micro skeleton and beginner ESP32 setup steps.	Added Code/tflite_micro_skeleton.cpp and expanded Code/README.md with ESP32 instructions
GitHub Copilot 12	Reported the TFLite Micro skeleton was incomplete.	Completed the skeleton with quantized support, Arduino setup, and a runnable loop
GitHub Copilot 13	Asked to avoid Roman-numbered pages in glossary references.	Explained how `\glsaddall` affects page references
GitHub Copilot 14	Asked to replace `iterrows` in the dataset loader with a better pattern.	Replaced loop with `Series.map` + boolean mask + `np.stack` for vectorized fingerprint loading
GitHub Copilot 15	Asked for help to deploy the quantized logP model onto an ESP32 with LiteRT.	Added a dedicated ESP-IDF project, a host-side Python serial script, and usage instructions for the ESP32 inference workflow
GitHub Copilot 16	Reported a serial port permission error while running the host SMILES sender against COM4.	Improved the Python script to catch serial open failures and print practical guidance about closing other serial monitors or processes
GitHub Copilot 17	Asked for a test pipeline that runs the dataset against the ESP32 logP model and produces charts	Added a resumable Python evaluation pipeline with subset selection modes, CSV summaries, and PDF plots
GitHub Copilot 18	Asked to reorganize the scripts for readability and maintainability.	Grouped imports, split constants into Paths and Hyperparameters sections, and reordered functions into Data, Model, Quantization and export, and Evaluation sections with separator comments
GitHub Copilot 19	Asked the ESP32 evaluator to retry failed molecules up to 100 times instead of moving on, and to treat device errors and timeouts as non-terminal so reruns retry them.	Added MAX_RETRIES constant, retry loop with serial buffer flush and re-PING between attempts, and changed load_completed_indices to only skip ok, invalid_smiles, and max_retries_exceeded results
GitHub Copilot 20	Asked why predicted logP never exceeds 5.466027.	Fixed by stratifying the calibration samples evenly across the sorted logP range.

System	Prompt	Usage
GitHub Copilot 21	Reported that clamping persisted at 5.480893 after recalibration and asked how to eliminate it entirely.	Changed output type to float32 (TFLite inserts a DEQUANTIZE op; Dense layers still run int8) and updated the firmware to register AddDequantize, check for float32 output, and read data.f[0] directly.
GitHub Copilot 23	Reported a boot loop after reflashing with the float32 output model.	Diagnosed peak arena overflow from the DEQUANTIZE op scratch buffers over-running the heap during Invoke(); doubled kTensorArenaSize from 64 KB to 128 KB.

Declaration of Authorship

I hereby confirm that I have personally and independently prepared the present work and have not used any sources or aids other than those specified. All passages taken verbatim or in substance from other sources are identified as such. The drawings, illustrations and tables in this work are created by me or have been provided with an appropriate source reference. This work has not been submitted by me to any other university in the same or similar form for the acquisition of an academic degree.

Frankfurt, March 29, 2026 ....

Rubin Chempananickal James